Enterolactone and breast cancer: methodological issues may contribute to conflicting results in observational studies.

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Lignans found in plant foods are converted by the intestinal microflora to enterolignans. The structure of enterolignans is similar to that of estrogens, which has inspired researchers to examine a potential protective association in relation to health outcomes. Numerous epidemiological studies have measured concentration of enterolignans, mainly enterolactone, in blood or urine as a biomarker of lignan exposure and studied its relation to breast cancer risk. Case-control studies have shown decreased breast cancer risk associated with high circulating enterolactone concentrations, but results demonstrated by prospective cohort studies are less clear. The purpose of this review is to discuss factors that may contribute to these contradictory findings obtained in epidemiological studies, including age distribution, enterolactone measurement error, heterogeneity of breast cancer subtypes, and genetic factors. Different sources of enterolactone precursors may also contribute to inconclusive results. In conclusion, to get robust evidence of the health effects of lignans and enterolactone, more effort has to be put on methodological problems, including reducing measurement errors in enterolactone estimation, and to identify factors that modify the effect.